

ANDY + DEAN • PRODUCT REVIEW

KitchenSmart prototype walkthrough

Every prototype page, in a practical meeting order. Review the product shape first; use the prompts to lock the decisions that unblock the next build phase.

Prototype v3

17 core views

6 edge states

18 Sep 2026

HOW TO USE THIS DECK

Start with the product story

The screens are sequenced like a real household journey, not by implementation order.

1 · Orient

Home → Inventory → Add Food. Confirm whether the dashboard is the right front door.

2 · Capture

Barcode match, no match, manual entry, and item corrections. Check whether provenance feels understandable.

3 · Decide safety

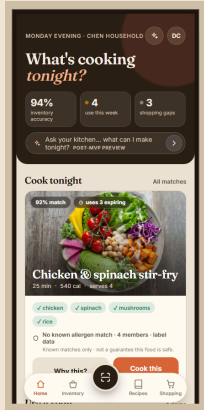
Allergies, profile visibility, recipe verdicts, and blocked states. These are policy decisions wearing UI clothes.

4 · Close the loop

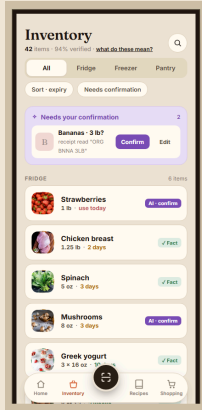
Recipes → cook confirmation → shopping → assistant. Confirm what belongs in MVP versus later.

Meeting rule: react to the decision and user flow, not typography polish. “Would we ship this behavior?” is the useful question.

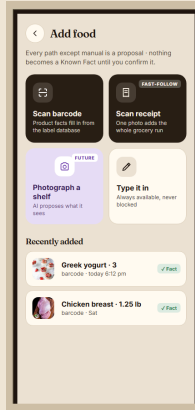
From “what do I have?” to trustworthy inventory



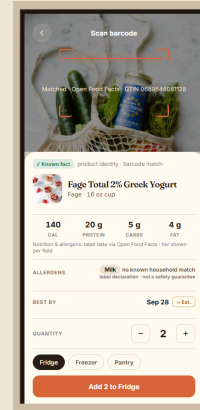
Home dashboard
Daily starting point



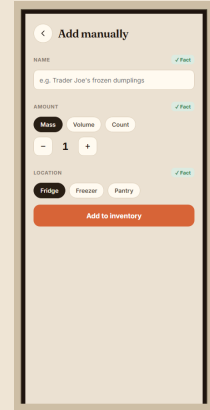
Inventory
What is on hand



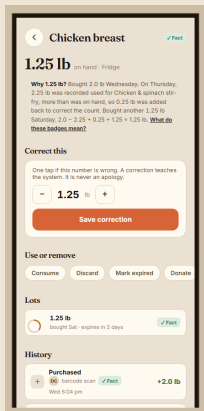
Add food
Choose capture method



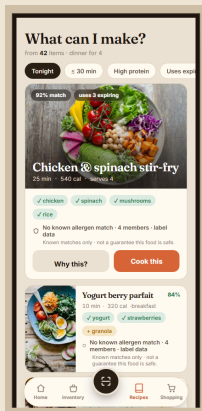
Barcode match
Review catalog facts



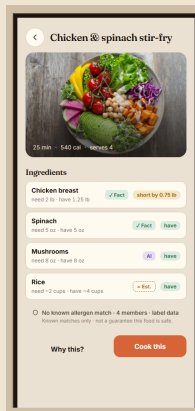
Manual add
Fallback capture



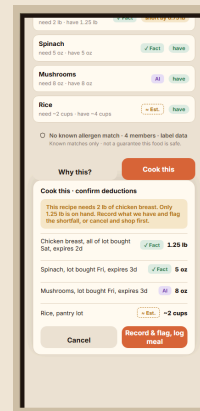
Item detail
Quantity + ledger



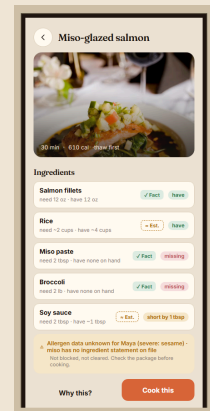
Recipes
Ranked meal ideas



Recipe · ready
Known match path

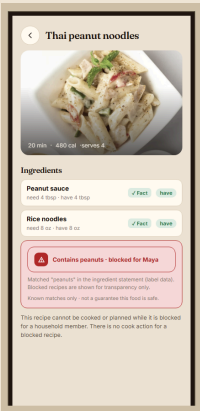


Cook confirm
Deductions + shortfall

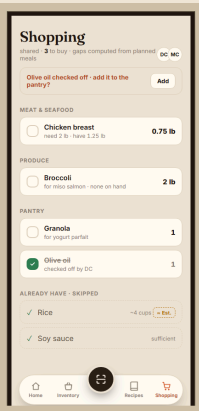


Recipe · unknown
Uncertainty path

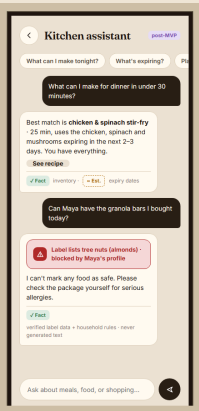
From meal choice to household safety



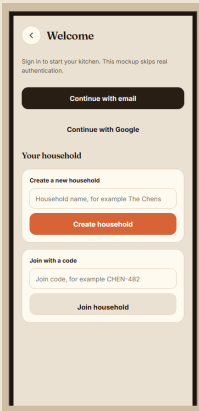
Recipe · blocked
Allergen stop



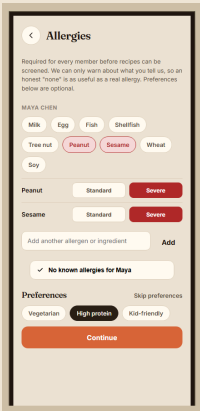
Shopping
Shared gaps



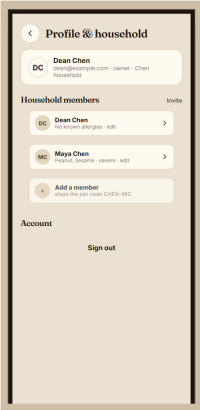
Assistant
Ask the kitchen



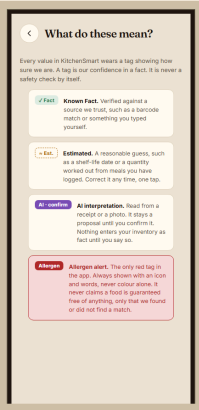
Household setup
Create or join



Allergies
Safety onboarding



Profile
Members + settings



Badge legend
Trust vocabulary

SCREEN 01 · HOME DASHBOARD

Daily starting point

S3 · Existing household

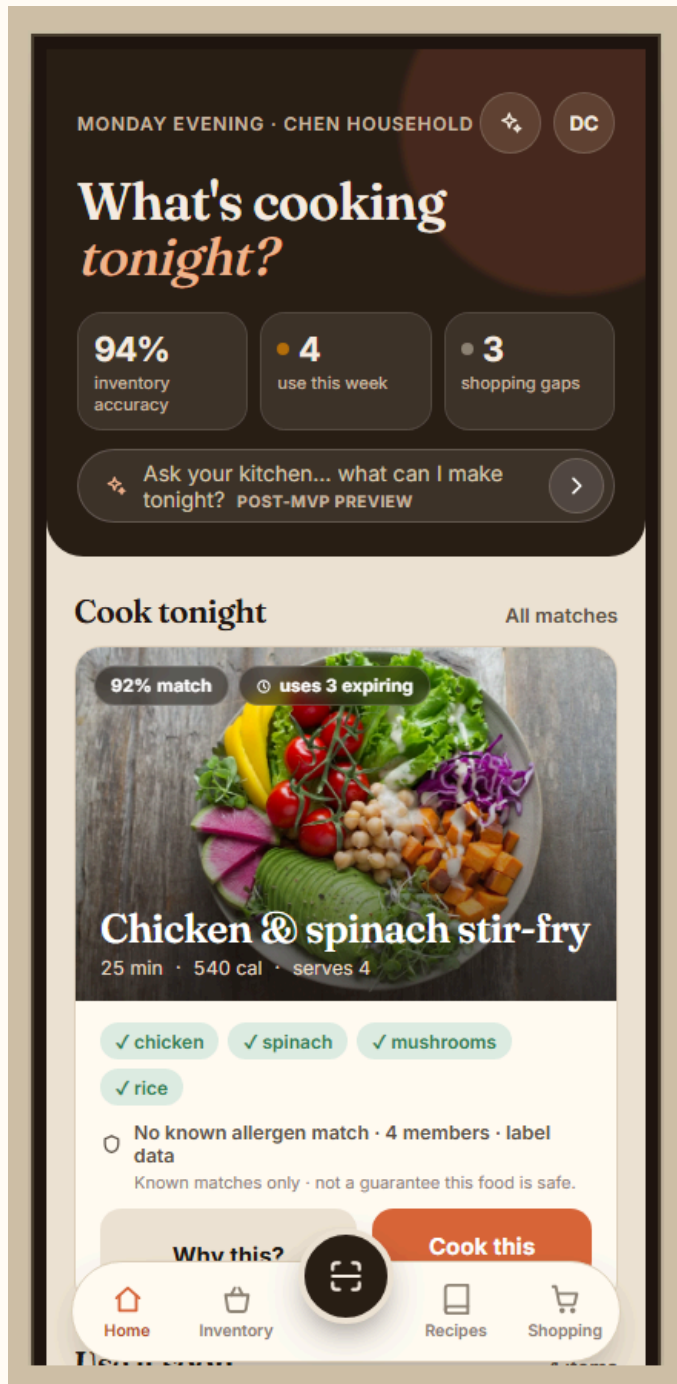
WHAT THIS PAGE IS DOING

- Makes expiring food and a strong dinner suggestion immediately useful.
- Provides fast routes to inventory, recipes, shopping, and the assistant.

Decision prompt · A1 · MVP scope / A10 · Home direction

Is the dashboard the right front door, or should the app open directly to Inventory?

FLOW · Home → Inventory / Recipes / Add / Assistant



SCREEN 02 • INVENTORY

What is on hand

S4 • Core MVP

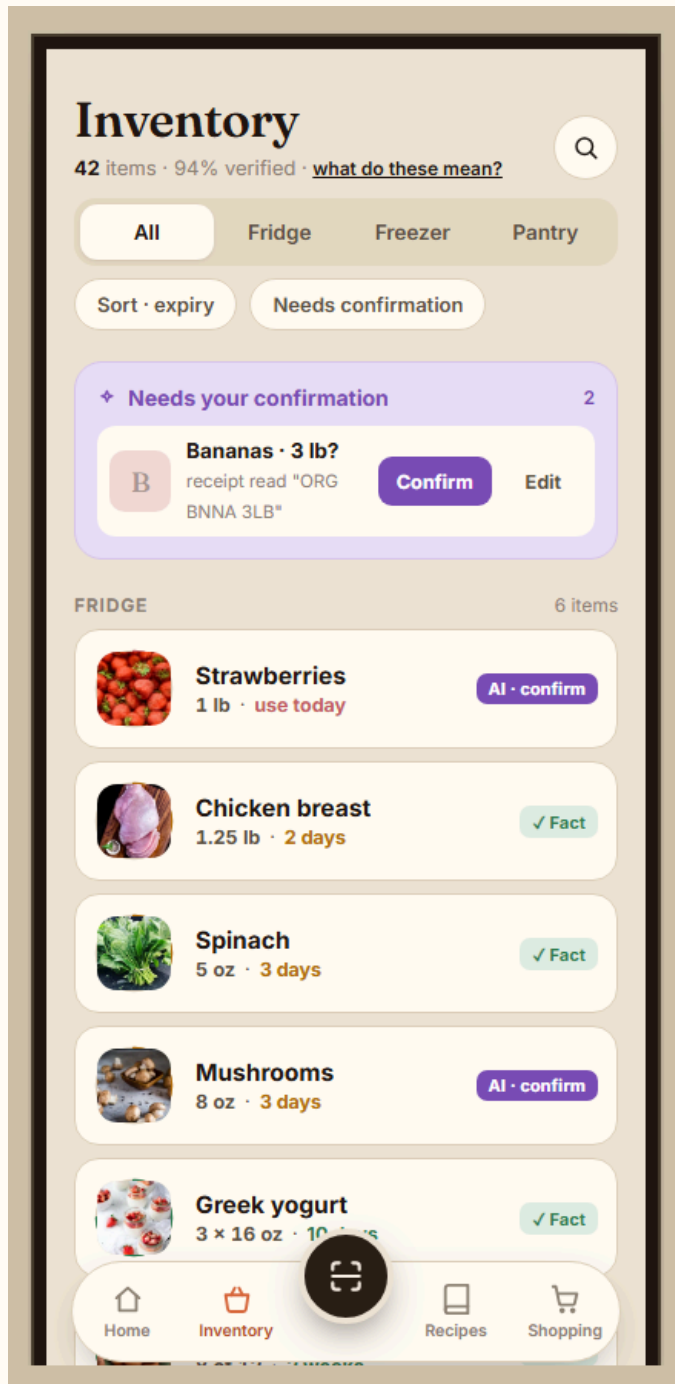
WHAT THIS PAGE IS DOING

- Location tabs keep fridge, freezer, and pantry understandable.
- Provenance and "needs confirmation" are visible before acting.

Decision prompt • A1 • MVP scope

Does this list expose enough information to trust quantities and expiry without opening every item?

FLOW • Home → Inventory → Item detail



Choose capture method

S6 · Capture hub

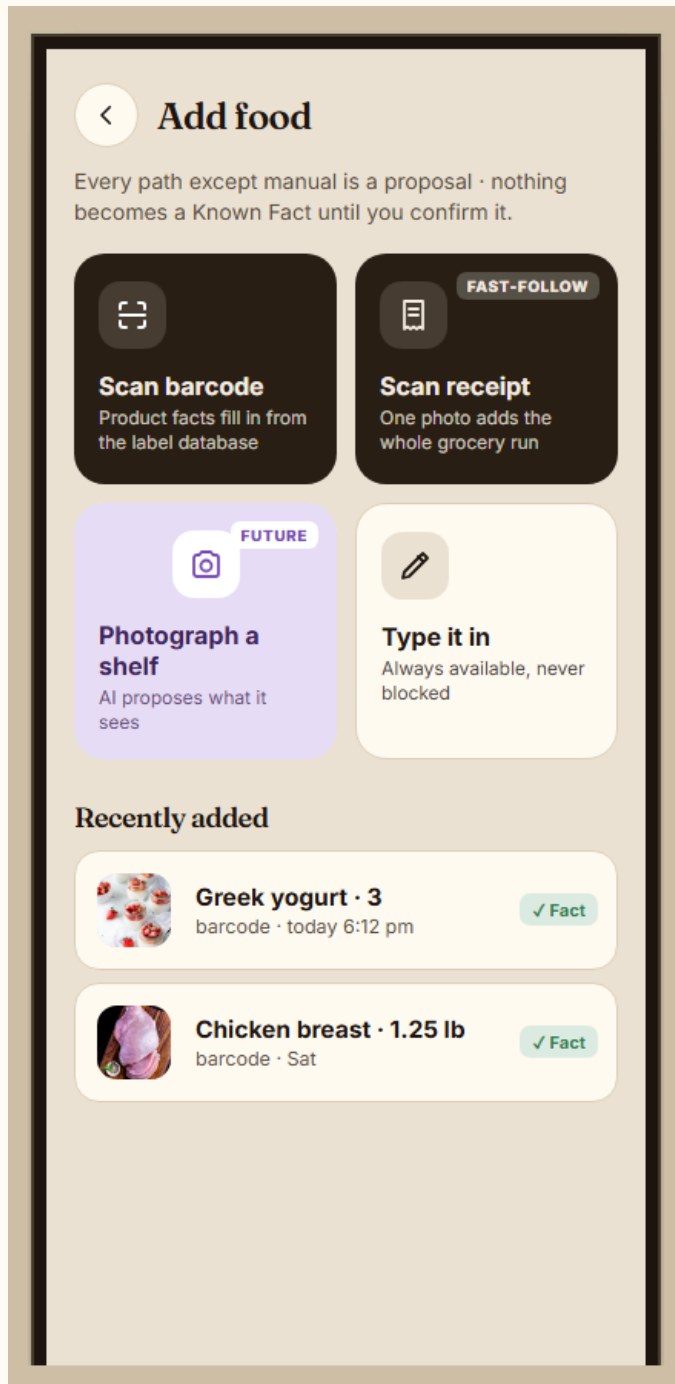
WHAT THIS PAGE IS DOING

- Sets honest expectations instead of pretending every capture method ships now.
- Keeps the primary action focused on getting food into inventory.

Decision prompt · A1 · Scope / A11 · Imagery

For the first release, are barcode scan and manual entry enough? Receipt and camera recognition are marked as later phases.

FLOW · Add → Barcode / Manual / Later capture



SCREEN 04 • BARCODE MATCH

Review catalog facts

S7 • Core MVP

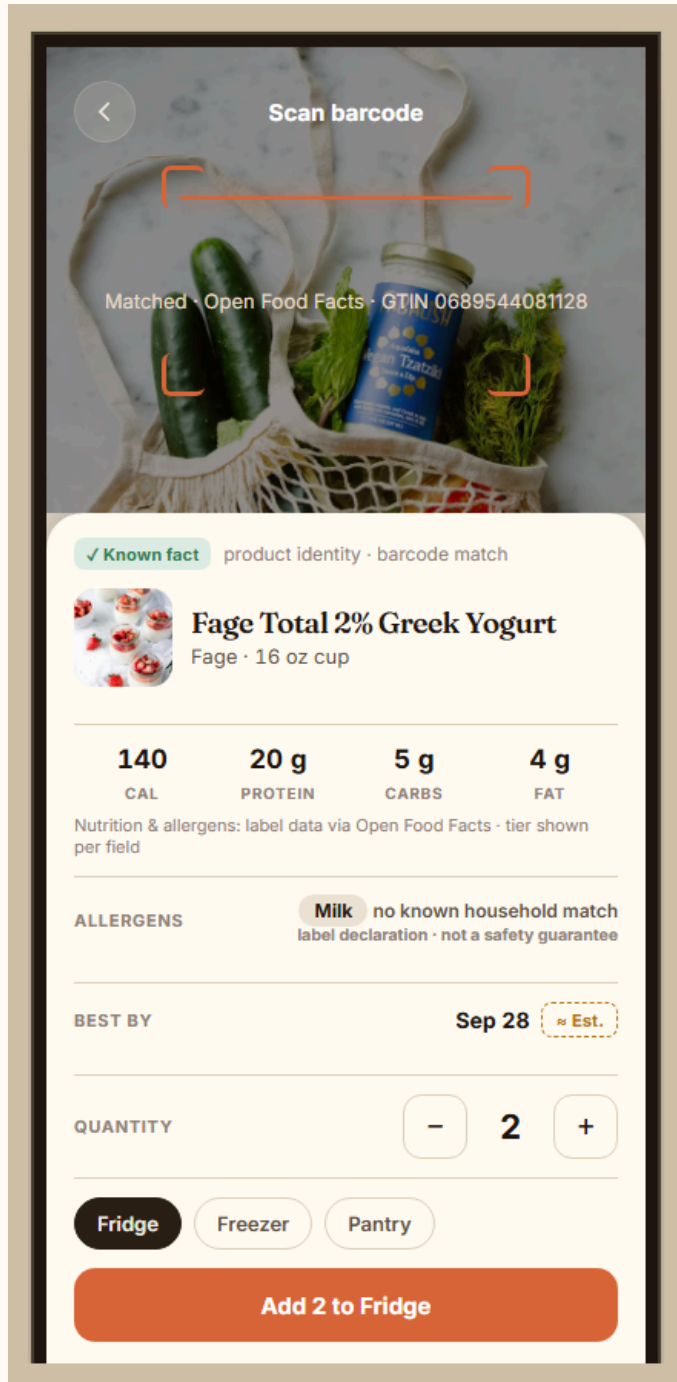
WHAT THIS PAGE IS DOING

- Shows source and GTIN, then asks the user to confirm quantity and location.
- Avoids silently turning third-party catalog data into a known fact.
- Who may promote label data to Known Fact is open (D-017); the prototype shows it as Estimated per field.

Decision prompt • A8 • Label-data trust tier

Should Open Food Facts label data appear as Estimated until the user confirms it?

FLOW • Scan → Review match → Add to inventory



Fallback capture

S9 • Core fallback

WHAT THIS PAGE IS DOING

- Unit type changes the available units.
- A missed barcode can be kept for later enrichment.

Decision prompt • A1 • MVP scope

Is the manual fallback fast enough for products with no barcode match and fresh produce?

FLOW • Add → Manual → Inventory

The image shows a mobile app interface for adding items manually. At the top, there's a back arrow and the title 'Add manually'. Below this, there are three sections: 'NAME', 'AMOUNT', and 'LOCATION'. Each section has a 'Fact' status indicator. The 'NAME' section has a text input field with the placeholder 'e.g. Trader Joe's frozen dumplings'. The 'AMOUNT' section has three unit type buttons: 'Mass' (selected), 'Volume', and 'Count'. Below these are three buttons: a minus sign, the number '1', and a plus sign. The 'LOCATION' section has three location buttons: 'Fridge' (selected), 'Freezer', and 'Pantry'. At the bottom, there is a large orange button labeled 'Add to inventory'.

Quantity + ledger

S5 · Core MVP

WHAT THIS PAGE IS DOING

- Quantity edits become append-only corrections, not silent overwrites.
- Used up, discarded, and donated require a reason.

Decision prompt · A9 · Shortfall / correction policy

Does the ledger explanation make a corrected quantity feel trustworthy rather than mysterious?

FLOW · Inventory → Item → Correct / consume

Chicken breast ✓ Fact

1.25 lb on hand · Fridge

Why 1.25 lb? Bought 2.0 lb Wednesday. On Thursday, 2.25 lb was recorded used for Chicken & spinach stir-fry, more than was on hand, so 0.25 lb was added back to correct the count. Bought another 1.25 lb Saturday. $2.0 - 2.25 + 0.25 + 1.25 = 1.25$ lb. What do these badges mean?

Correct this

One tap if this number is wrong. A correction teaches the system. It is never an apology.

- **1.25** lb +

Save correction

Use or remove

Consume Discard Mark expired Donate

Lots

1.25 lb ✓ Fact
bought Sat · expires in 2 days

History

Purchased
+ DC barcode scan ✓ Fact **+2.0 lb**
Wed 6:04 pm

Ranked meal ideas

S10 • Recommended scope

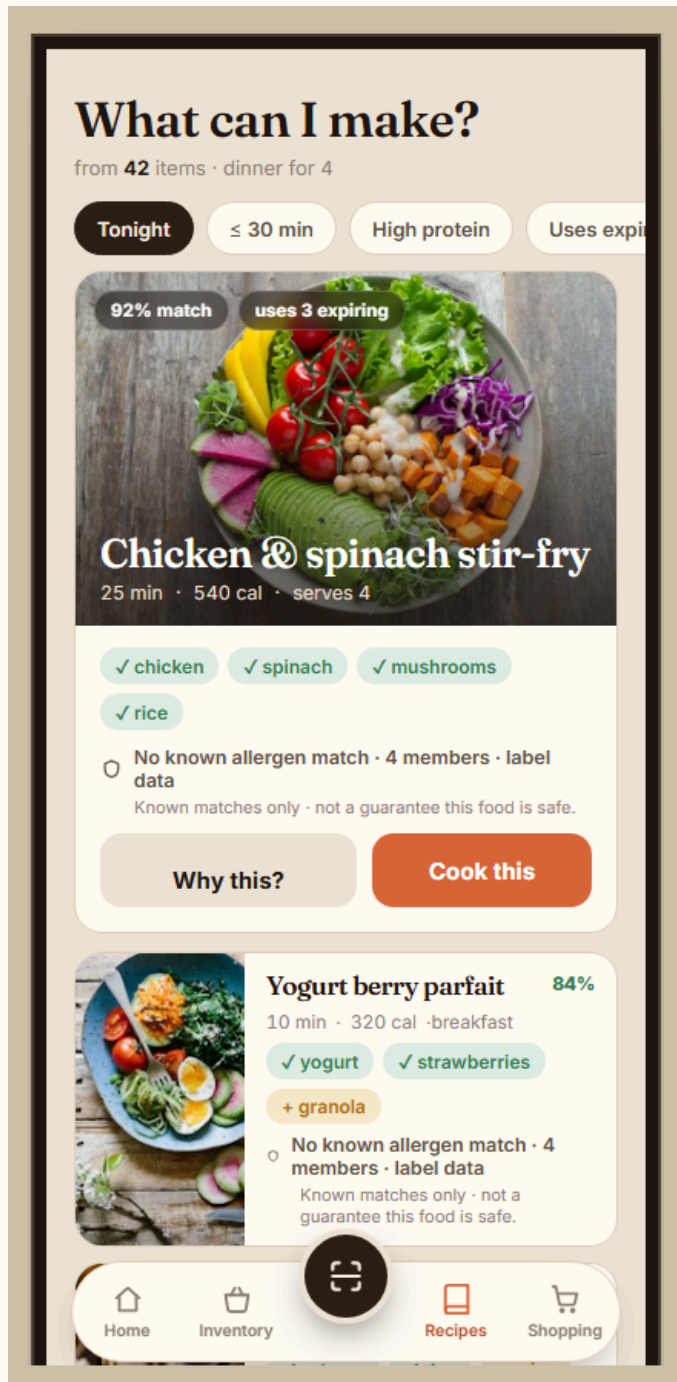
WHAT THIS PAGE IS DOING

- Ranking explains expiry usage, purchase gaps, time, and preferences.
- Safety verdicts are separate from the ranking score.

Decision prompt • A1 • MVP scope

Are ranked recipe suggestions essential for MVP, or should the first slice stop at trustworthy inventory?

FLOW • Home → Recipes → Recipe detail



Known match path

S11 • Recipe detail

WHAT THIS PAGE IS DOING

- Shows exact on-hand amounts and the evidence tier.
- Cook action leads to explicit inventory deductions.

Decision prompt • A8 • Trust tier

Is “no known allergen match” clear enough without sounding like a safety guarantee?

FLOW • Recipes → Ready recipe → Cook

<

Chicken & spinach stir-fry

25 min · 540 cal · serves 4

Ingredients

Chicken breast

need 2 lb · have 1.25 lb

✓ Fact

short by 0.75 lb

Spinach

need 5 oz · have 5 oz

✓ Fact

have

Mushrooms

need 8 oz · have 8 oz

AI

have

Rice

need ~2 cups · have ~4 cups

≈ Est.

have

No known allergen match · 4 members · label data

Known matches only · not a guarantee this food is safe.

Why this?

Cook this

SCREEN 09 · COOK CONFIRM

Deductions + shortfall

S11 · Cook confirm

WHAT THIS PAGE IS DOING

- Every deduction names the lot it comes from, first expiry first.
- The shortfall is written as a flagged correction, never a silent overwrite.
- Proposed policy (OQ-1): record the statement, flag the gap, then prompt to fix inventory.

Decision prompt · A9 · Shortfall policy

When inventory is short, should we record what exists, flag the gap, and still log the meal?

FLOW · Recipe → Cook this → Confirm deductions → Inventory

The screenshot shows a mobile app interface for confirming a meal. At the top, it lists ingredients with their required and available amounts: Chicken breast (need 2 lb, have 1.25 lb, short by 0.75 lb), Spinach (need 5 oz, have 5 oz), Mushrooms (need 8 oz, have 8 oz), and Rice (need ~2 cups, have ~4 cups). Below this, there's a section for allergen information. The main part of the screen is a 'Cook this · confirm deductions' dialog box. It contains a yellow warning box about the chicken breast shortfall. Below the warning, it lists the ingredients again with their lot and expiry dates. At the bottom of the dialog are two buttons: 'Cancel' and 'Record & flag, log meal'.

need 2 lb · have 1.25 lb ✓ Fact short by 0.75 lb

Spinach
need 5 oz · have 5 oz ✓ Fact have

Mushrooms
need 8 oz · have 8 oz AI have

Rice
need ~2 cups · have ~4 cups ≈ Est. have

☐ No known allergen match · 4 members · label data
Known matches only · not a guarantee this food is safe.

Why this? **Cook this**

Cook this · confirm deductions

This recipe needs 2 lb of chicken breast. Only 1.25 lb is on hand. Record what we have and flag the shortfall, or cancel and shop first.

Chicken breast, all of lot bought Sat, expires 2d ✓ Fact **1.25 lb**

Spinach, lot bought Fri, expires 3d ✓ Fact **5 oz**

Mushrooms, lot bought Fri, expires 3d AI **8 oz**

Rice, pantry lot ≈ Est. **~2 cups**

Cancel **Record & flag, log meal**

Uncertainty path

S11 • Uncertainty path

WHAT THIS PAGE IS DOING

- Names the missing allergen data instead of hiding uncertainty.
- Unknown is a third state: no red, no green, cooking stays allowed with a warning.
- Separates shopping gaps from allergen unknowns.

Decision prompt • A8 • Label-data trust tier / D-017

When an ingredient has no ingredient statement on file, is "not blocked, not cleared" plus a check-the-package prompt the right answer?

FLOW • Recipes → Unknown recipe → Check package → Cook

<
Miso-glazed salmon



30 min · 610 cal · thaw first

Ingredients

Salmon fillets
 need 12 oz · have 12 oz
 ✓ Fact
have

Rice
 need ~2 cups · have ~4 cups
 ≈ Est.
have

Miso paste
 need 2 tbsp · have none on hand
 ✓ Fact
missing

Broccoli
 need 2 lb · have none on hand
 ✓ Fact
missing

Soy sauce
 need 2 tbsp · have ~1 tbsp
 ≈ Est.
short by 1 tbsp

⚠ Allergen data unknown for Maya (severe: sesame) · miso has no ingredient statement on file

Not blocked, not cleared. Check the package before cooking.

Why this?
Cook this

Allergen stop

S11 • Safety stop

WHAT THIS PAGE IS DOING

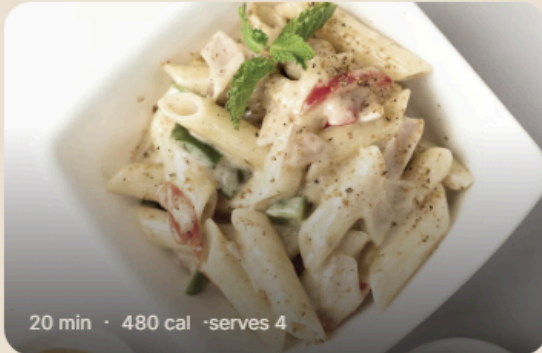
- Allergen red is reserved for an actual known match.
- The reason and affected household member are named.

Decision prompt • A3 / A8 • Household safety policy

Should blocked recipes remain visible for transparency, with no cook action?

FLOW • Recipes → Blocked detail → No cook action

< Thai peanut noodles



20 min • 480 cal • serves 4

Ingredients

Peanut sauce

need 4 tbsp • have 4 tbsp

✓ Fact

have

Rice noodles

need 8 oz • have 8 oz

✓ Fact

have



Contains peanuts • blocked for Maya

Matched "peanuts" in the ingredient statement (label data).
Blocked recipes are shown for transparency only.

Known matches only • not a guarantee this food is safe.

This recipe cannot be cooked or planned while it is blocked for a household member. There is no cook action for a blocked recipe.

SCREEN 12 • SHOPPING

Shared gaps

S13 • Shared utility

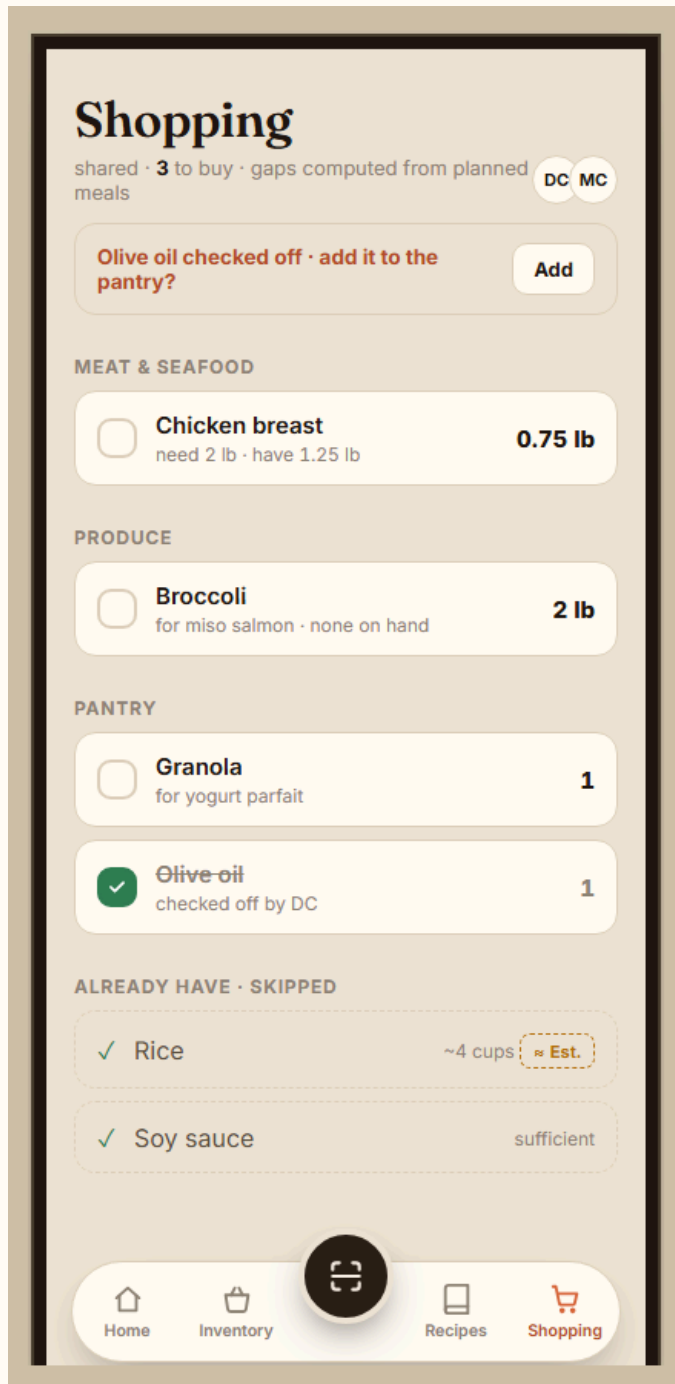
WHAT THIS PAGE IS DOING

- Meal-plan gaps become a plain shared list.
- Offline actions queue rather than disappear.

Decision prompt • A1 • MVP scope

Does shared shopping belong in the reduced vertical slice, or can it follow inventory and recipes?

FLOW • Recipes / Assistant → Shopping



Ask the kitchen

S14 • AI surface

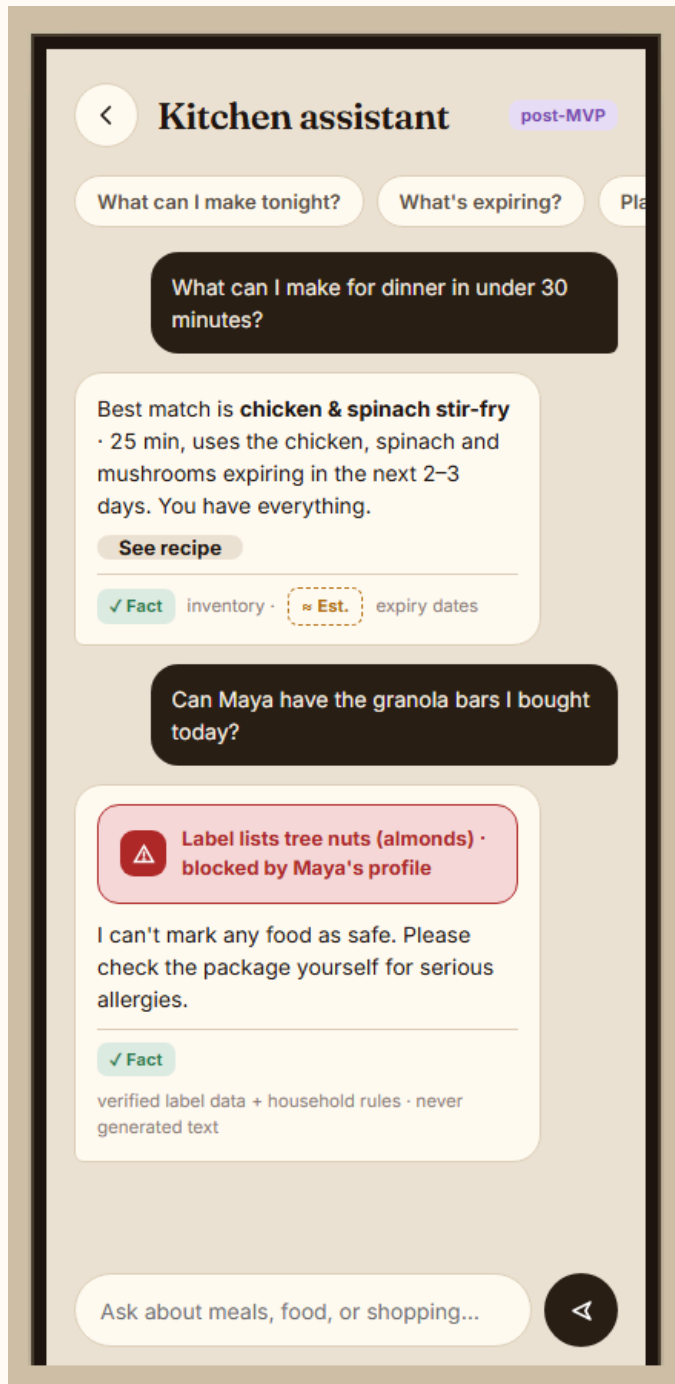
WHAT THIS PAGE IS DOING

- Prompt starters make the feature concrete.
- The assistant should explain and act through deterministic app functions.

Decision prompt • A1 • MVP scope / B5 • LLM cost

What is the one assistant job worth shipping first: meal ideas, expiry help, or list building?

FLOW • Home → Assistant → Recipe / Shopping action



SCREEN 14 · HOUSEHOLD SETUP

Create or join

S1 · New household

WHAT THIS PAGE IS DOING

- Supports create and join without assuming a single-person account.
- This is where owner/member rules first become visible.

Decision prompt · A3 · Household permissions

Who may create, join, invite, and manage a household in the first release?

FLOW · Sign in → Create or join → Allergies

The mockup shows a mobile app interface for household setup. At the top, there's a back arrow and the title 'Welcome'. Below this is a message: 'Sign in to start your kitchen. This mockup skips real authentication.' There are two main buttons: 'Continue with email' (dark brown) and 'Continue with Google' (light brown). Underneath is the section 'Your household'. It contains two sub-sections: 'Create a new household' and 'Join with a code'. The 'Create a new household' section has a text input field with placeholder text 'Household name, for example The Chens' and a dark brown 'Create household' button. The 'Join with a code' section has a text input field with placeholder text 'Join code, for example CHEN-482' and a light brown 'Join household' button.

SCREEN 15 • ALLERGIES

Safety onboarding

S2 • Safety onboarding

WHAT THIS PAGE IS DOING

- Severity is explicit per allergen.
- Preferences are kept visually separate from safety constraints.

Decision prompt • A2 / A3 / A10 • Market + permissions

Confirm: US nine allergens, owner enters them for all members, and the household can see allergy flags.

FLOW • Household setup → Allergies → Home

The screen is titled "Allergies" with a back arrow. It contains a paragraph explaining the requirement for allergy screening. Below this, the name "MAYA CHEN" is displayed. A grid of allergen buttons includes Milk, Egg, Fish, Shellfish, Tree nut, Peanut, Sesame, Wheat, and Soy. Peanut and Sesame are highlighted in red. Below the grid, there are two rows for severity selection: "Peanut" with "Standard" and "Severe" options, and "Sesame" with "Standard" and "Severe" options. A text input field "Add another allergen or ingredient" with an "Add" button is present. A confirmation message "✓ No known allergies for Maya" is shown. At the bottom, there is a "Preferences" section with "Skip preferences" link, and three buttons: "Vegetarian", "High protein" (selected), and "Kid-friendly". A large orange "Continue" button is at the very bottom.

< Allergies

Required for every member before recipes can be screened. We can only warn about what you tell us, so an honest "none" is as useful as a real allergy. Preferences below are optional.

MAYA CHEN

Milk Egg Fish Shellfish

Tree nut Peanut Sesame Wheat

Soy

Peanut Standard Severe

Sesame Standard Severe

Add another allergen or ingredient Add

✓ No known allergies for Maya

Preferences Skip preferences

Vegetarian High protein Kid-friendly

Continue

SCREEN 16 • PROFILE

Members + settings

S12 • Household profile

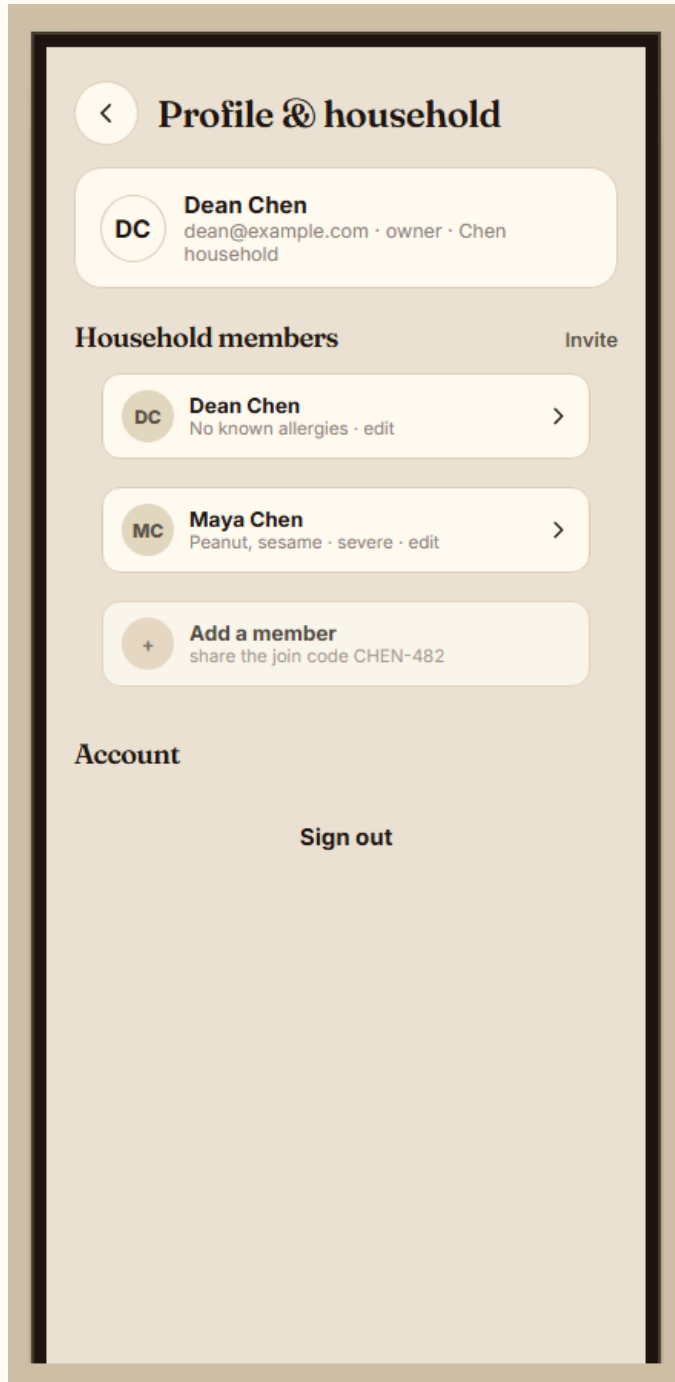
WHAT THIS PAGE IS DOING

- Central place for people, safety profiles, and household settings.
- Name choice must happen before store/domain work.

Decision prompt • A3 / A10 • Privacy + product name

Which member details are household-visible, and is “KitchenSmart” only a placeholder?

FLOW • Avatar → Household profile → Member allergy edit



Trust vocabulary

System explanation

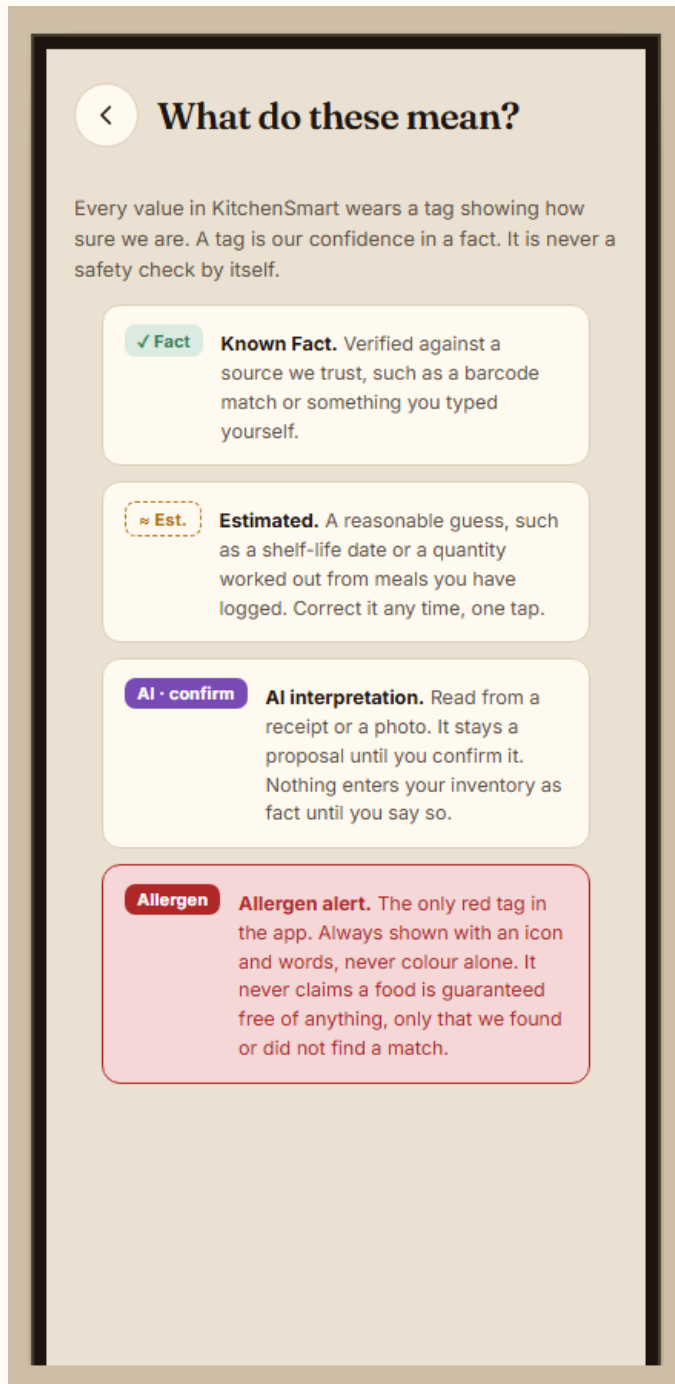
WHAT THIS PAGE IS DOING

- Shape, icon, label, and color all carry meaning.
- Review these colors on a real phone before adopting the proposed contrast fixes.

Decision prompt · A8 / A12 · Trust + contrast

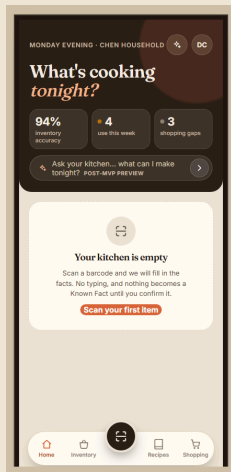
Do Known Fact, Estimated, and AI Interpretation feel distinct and understandable without relying on color alone?

FLOW · Any provenance badge → Legend



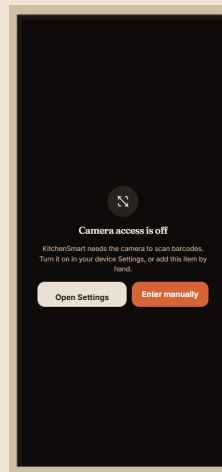
Don't approve only the happy path

These states usually reveal more product truth than the polished default screens.



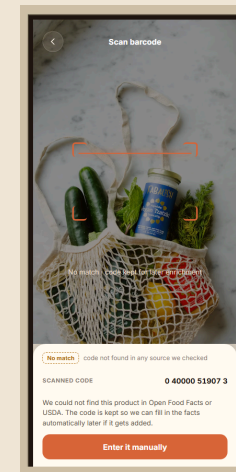
New user / empty home

First-run value before inventory exists



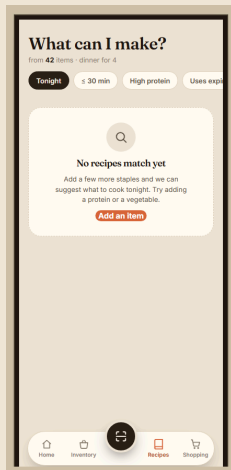
Camera denied

Recovery path to settings or manual entry



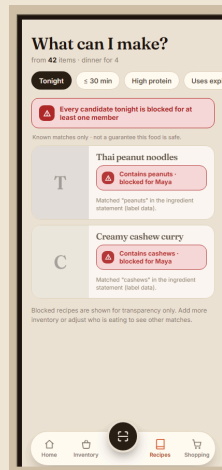
No barcode match

Keep code and fall back to manual



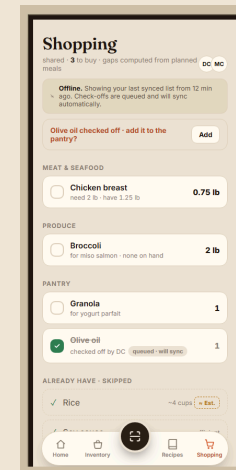
No recipe results

Explain why and offer a useful next move



Blocked recipe list

Transparent but clearly unavailable



Shopping offline

Queue shared changes for sync

CLOSE THE MEETING

Decisions to leave with

MVP shape

Ratify the reduced slice, or name the one or two additions that are truly non-negotiable.

Household safety

Confirm US-first allergens, owner/member rights, visibility, and label-data trust.

Product direction

Keep dashboard home? Light-first? Is KitchenSmart just a working name?

Build path

Expo client, Better Auth, stubbed-auth M2 now, and hosting only when shared testing needs it.

Afterward: record the answers in DECISIONS.md. Until then, every highlighted item here is a proposal, not a settled requirement.